

BRYAN CRUZ

SENIOR FULL STACK ENGINEER | AI/ML SPECIALIST

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SUMMARY

Senior Full-Stack Engineer with a proven track record of building scalable EdTech solutions and integrating advanced features with generative AI models (LLMs, RAG, Flowise). Expert in front-end ecosystems (React, Angular), enterprise back-end frameworks (Java/Spring Boot, Node.js/Next.js), and PHP platforms (Moodle, WordPress).

Adept at technical leadership and driving end-to-end project lifecycles, leveraging containerized infrastructure (Docker) and cloud platforms to orchestrate modern deployments.

Currently pursuing a Master's degree in Software Engineering, expanding my expertise into microservices architectures (Java, Spring Cloud) and the implementation of cybersecurity practices (CWE, Pentesting) for the development of robust and secure software.

Seeking high-impact roles in AI-driven architecture or senior full-stack engineering.

SKILLS

Languages: JavaScript (ES6+) • TypeScript • PHP • Java • SQL

Frontend: ReactJS • Next.js • Angular 2+ • Zustand • Tailwind CSS • HTML5/CSS3

Backend & Cloud: Node.js • GraphQL • REST API • Java • Spring Boot • Netflix Eureka • API Gateway • AWS (EC2, S3, RDS) • Docker • CI/CD (GitLab, Jenkins) • ElasticSearch

AI & LLMs: Flowise • RAG • Fine-tuning • Vector DB • LangChain • Computer Vision (YOLO) • Prompt Engineering

Cybersecurity: Vulnerability Analysis (CWE), Penetration Testing, Parrot Security OS, Burp Suite, Secure Authentication (Ghost JWT).

Testing & QA: PHPUnit, Behat (BDD), Jest, Cypress, TDD, E2E Testing

Databases: MySQL • PostgreSQL • MongoDB • Redis • Vector Database (Pinecone)

Other: English (Advanced C1), Spanish (Native), Portuguese (Basic).

WORK EXPERIENCE

Full Stack Software Engineer & Generative AI

Open LMS (General Atlantic Portfolio - US Remote)

Feb 2025 - Present

- **AI Project Leadership:** Led the end-to-end product architecture of enterprise-grade AI plugins, radically streamlining content creation workflows for an ecosystem of more than 8 million users in over 120 countries.
- **AI Solution Development:** Architected and shipped [tiny_ai_assistant](#) (AI content creator for Moodle) and an automated question generator using files and prompt engineering, enabling learning materials to be generated with high user satisfaction and significant time savings for educators.
- **Quality Automation (DevOps + AI):** Implemented Gitlab pipelines that incorporate Artificial Intelligence routines for reviewing and validating potential errors in unit and automated tests, raising the quality of the code and the CI/CD process.

Associate Software Engineer

Learning technologies Group (UK Remote)

Mar 2022 - Feb 2025

- **Lead Refactoring:** Led the architectural modernization of the Snap theme for Moodle 4.x, aggressively eradicating technical debt and optimizing rendering performance to support thousands of concurrent global users. See [Theme Snap contributions](#)
- **Reactive UI Architecture:** Redesigned complex interfaces using Angular and advanced reactive state management patterns (ReactJS principles), driving the development of highly modular user experiences.
- **System Optimization & Reliability:** Optimized critical enterprise modules and legacy plugins. Engineered robust web services, tuned MySQL database queries, and implemented testing practices to ensure platform stability and high availability.

WordPress Consultant & Automations

Freelancer (Colombia)

Jun 2021 - Aug 2023

- **Scalable Web Architecture:** Architected and scaled digital presence for SMEs, optimizing database performance and security for WooCommerce environments, resulting in faster load times and increased conversion rates
- **Artificial Intelligence (AI) Integration:** Implement generative AI solutions for the efficient creation of content assets, optimizing clients' digital strategy (ad variations using AI, WhatsApp automations, Social media design).
- **Analysis and Data:** Implementation of analytics and metrics management tools (Meta Ads) for data-driven decision-making and optimization by objectives (traffic, leads).

Artificial Intelligence Research Intern

Universidad del Rosario · Apprenticeship

Jan 2021 - Aug 2021

- Develop cutting-edge research in Deep Learning, implementing the YOLO (You Only Look Once) model with Python and Google Colab for computer vision applications in monitoring and evaluating people.
- **International Publication:** Authored and presented research at the 17th International Symposium on Medical Information Processing in Brazil, validating technical findings at a global academic level.
- I contributed to the university's AI research group in multiple projects and debates, strengthening my theoretical and practical understanding of the fundamentals of Artificial Intelligence, which I now apply in the development of plugins for Moodle.

PROJECTS, AWARDS & PUBLICATIONS

Distributed Microservices Platform for Online Library

International University of Rioja - Spain

2026

Engineered a high-availability backend architecture leveraging Java and Spring Boot. Implemented service discovery via Netflix Eureka and an API Gateway utilizing round-robin load balancing. Integrated Elasticsearch for high-performance indexing and built a responsive ReactJS frontend. See [ReactJS frontend](#) and [RelatosPapelBackend](#)

Publication: Monitoring and evaluation of people indoors and outdoors using deep learning

Rosario University - Colombia

2021

Developed a real-time computer vision pipeline for human activity monitoring using Python, YOLO, and Deep Learning architectures (TensorFlow/PyTorch). Engineered the object detection and human pose estimation models to process low-cost video streams in real-time, delivering an accessible e-health solution for behavior analysis.

National Recognition - "Ser Pilo Paga 2"

Colombian government

2016 - 2021

Excellence Scholarship for achieving the highest score on the national standardized tests (ICFES).

EDUCATION / CERTIFICATIONS

Generative AI Foundations

Certiport, a Pearson VUE business (2026)

Master of Science (M.Sc.) in Software Engineering

International University of Rioja. - Spain (2025 - Present)

Advanced Program in Generative AI

International University of Rioja. - Spain (2025 - Present)

Fullstack Web Development Bootcamp

UT Talento Tech - Cymetria (2024)

Biomedical Engineering

Colombian School of Engineering Julio Garavito & Rosario University - Colombia (2016 - 2021)